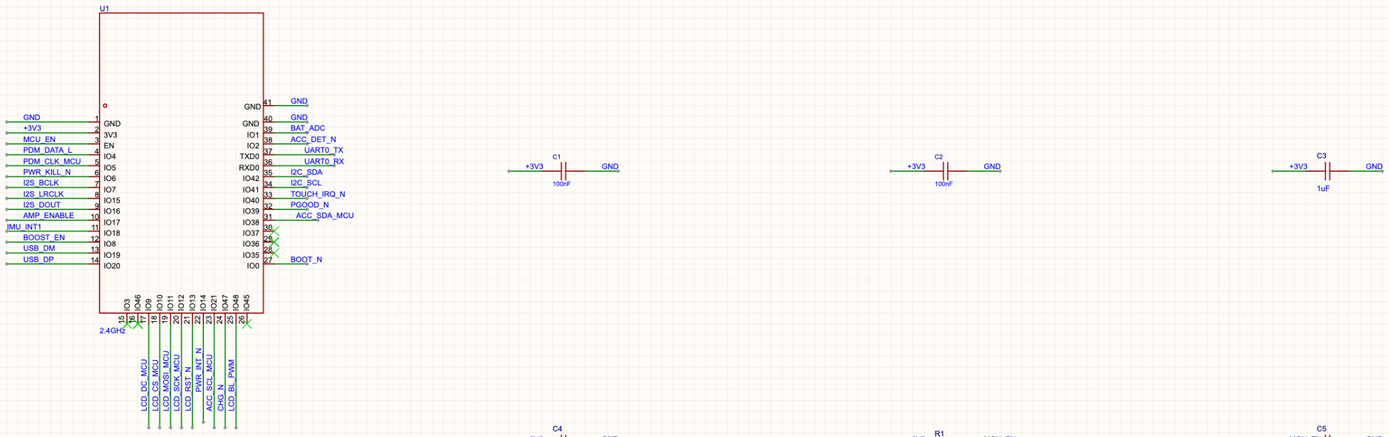
SPIRIT / 01_MCU / 原理图复核版 2026-10-18



POWER: GPIO14 = active-low shutdown interrupt; GPIO6 = open-drain KILL; GPIO44 RX / GPIO43 TX test pads. Keep BOOT / RESET for recovery.

SPIRIT / 02_I2C / 原理图复核版 2026-10-18

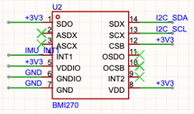
ngspice简化模型：200pF上升时间约796ns；400kHz不通过300ns上升时间要求。

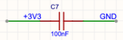
I2C默认100kHz：板上4.7k上拉，目标总线电容≤200pF。外接模块不得重复并联过低上拉。

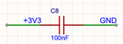


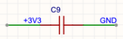


SPIRIT / 03_IMU / 原理图复核版 2026-10-18

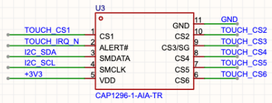


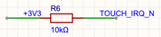






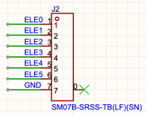
SPIRIT / 04_TOUCH / 原理图复核版 2026-10-18















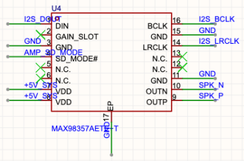








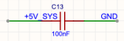
SPIRIT / 05_AUDIO / 原理图复核版 2026-10-18

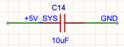






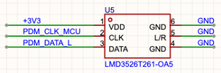




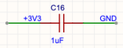




SPIRIT / 06_MIC / 原理图复核版 2026-10-18











原理图	Spirit_Native	创建日期	2026-10-24
		更新日期	2026-10-24
板子	01_主控板_Main	图页	06_MIC
绘制			
审阅		Spirit_V080_Three_Board_Layout_20261026	
		版本	尺寸
嘉立创EDA		V1.0	A2

嘉立创EDA

SPIRIT / GPIO直接连接分配 / V074F 2026-10-23

GPIO47 / module24 = CHG_N; GPIO39 / module32 = PGOOD_N. R114/R115 pull-ups to 3V3 retained.

PGOOD LOW = valid USB input. This is not physical dock detection; J4 dock contact removed by user.

J6 auxiliary key removed; SW4 POWER remains. USB voltage detector Q104/R105/R106/R107 removed.

N16R8: GPIO35/36/37 reserved by octal PSRAM; GPIO3/45/46 kept NC (strapping); GPIO43/44 reserved UART TX/RX test pads.

36 exposed GPIO: 3 PSRAM reserved, 28 functional incl. BOOT, 2 UART test pads, 3 unassigned strap pins.

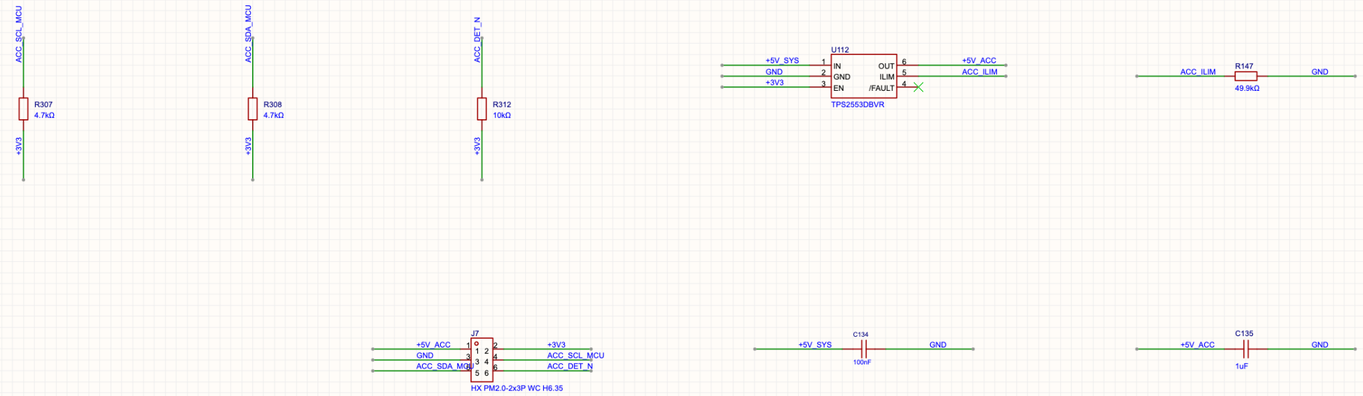
GPIO14=PWR_INT_N; GPIO6=PWR_KILL_N open-drain. GPIO44=RX/TP114; GPIO43=TX/TP115. USB GPIO19/20 and BOOT/RESET retained.

CHG high is not a guaranteed 100% battery indication. Recharge/fault/PGOOD state must be handled per BQ24075 datasheet.

IO4	pad 04	PDM_DATA_L
IO5	pad 05	PDM_CLK_MCU
IO6	pad 06	PWR_KILL_N
IO7	pad 07	I2S_BCLK
IO15	pad 08	I2S_LRCLK
IO16	pad 09	I2S_DOUT
IO17	pad 10	AMP_ENABLE
IO18	pad 11	IMU_INT1
IO8	pad 12	BOOST_EN
IO19	pad 13	USB_DM
IO20	pad 14	USB_DP
IO3	pad 15	STRAP / NC
IO46	pad 16	STRAP / NC
IO9	pad 17	LCD_DC_MCU
IO10	pad 18	LCD_CS_MCU
IO11	pad 19	LCD_MOSI_MCU
IO12	pad 20	LCD_SCK_MCU
IO13	pad 21	LCD_RST_N
IO14	pad 22	PWR_INT_N
IO21	pad 23	ACC_SCL_MCU
IO47	pad 24	CHG_N
IO48	pad 25	LCD_BL_PWM
IO45	pad 26	STRAP / NC
IO0	pad 27	BOOT_N
IO35	pad 28	PSRAM RESERVED
IO36	pad 29	PSRAM RESERVED
IO37	pad 30	PSRAM RESERVED
IO38	pad 31	ACC_SDA_MCU
IO39	pad 32	PGOOD_N
IO40	pad 33	TOUCH_IRQ_N
IO41	pad 34	I2C_SCL
IO42	pad 35	I2C_SDA
RXD0	pad 36	UART0_RX / TP114
TXD0	pad 37	UART0_TX / TP115
IO2	pad 38	ACC_DET_N
IO1	pad 39	BAT_ADC

U7、J4、J6及专用外围已取消；BQ24075的CHG/PGOOD直接接主控，无GPIO扩展器。

08_ACCESSORY / 6-contact identifiable accessory interface / blade adapter V0.1



J7: 1=+5V_ACC through U112 current limiter; 2=+3V3 logic; 3=GND; 4=SCL; 5=SDA; 6=DET_N.

配件 DET 接 GND；I2C 为 3.3V 逻辑，默认 100kHz；上拉仅保留主板这一组。

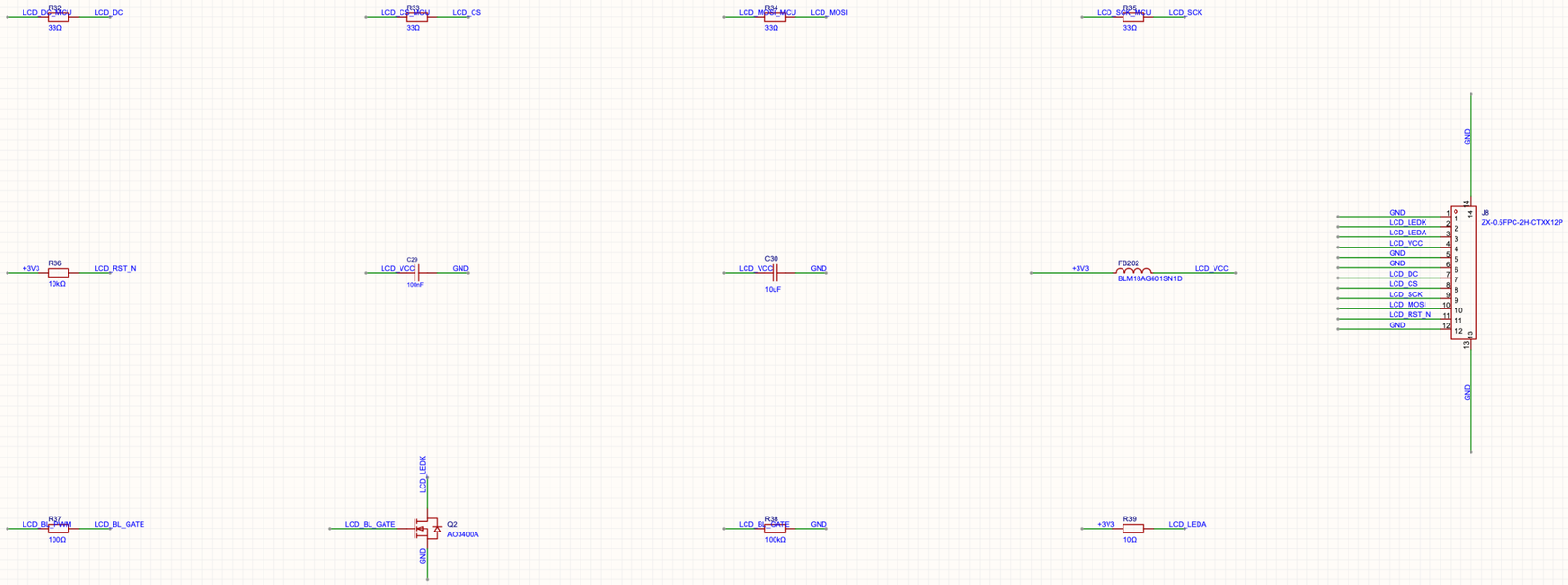
U112 TPS2553 / R147=49.9k limits 5V accessory supply near 0.52A typical. Firmware budget <=400mA; adapter logic <=50mA. Adapter U702 gates LEDs OFF until valid configuration and arm.

I2C 100kHz / 0x32 adapter identity; type 0x0001 = blade. Keyed power-off mating for prototype. Do not assume arbitrary hot-plug, external power input or high-power blade support.

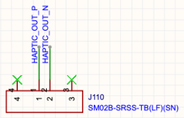
SPIRIT / 09_DISPLAY / 原理图复核版 2026-10-18

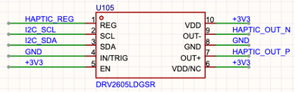
Single LCD / Usaki reference: +3V3 -> FB202 -> LCD_VCC; C29/C30 on LCD side near J8. BL: +3V3 -> R39 -> LEDA; LEDK -> Q2 PWM sink.

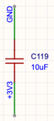
Rail is nominal 3.3V. R39=10 ohm provisional current limiting (not 0 ohm); verify LED current/brightness on actual LCD. Keep BL return away from IMU/microphone.

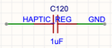


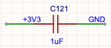
SPIRIT / 16_Motor / 原理图复核版 2026-10-18











V074F: DRV2605L EN -> switched +3V3; R135/R136 removed. Use I2C GO/STANDBY (power-up standby). No independent GPIO disable. GPIO6 now PWR_KILL_N.

原理图	Spirit_Native	创建日期	2026-10-23
		更新日期	2026-10-23
板子	01_主控板_Main	图页	16_Motor
绘制		Spirit_V080_Three_Board_Layout_20261026	
审阅			
		版本	尺寸
嘉立创EDA		V1.0	A2

SPIRIT / 17_Test_points / 原理图复核版 2026-10-18

V074F UART service pads (bare copper, no connector / no BOM component)

TP114 = UART0_RX / GPIO44 / module pad36; TP115 = UART0_TX / GPIO43 / module pad37.

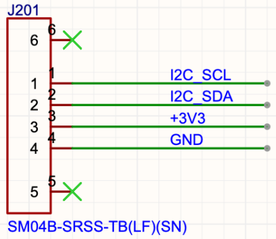
Use 3.3V UART logic, cross TX/RX and connect GND (TP113). Power board first. USB GPIO19/20 remains the normal download/debug path.

TP105 -> VSYS_RAW; TP107 -> +5V_SYS. Obsolete MAIN_IN / BOOST_5V test-pad net names corrected to match V074D power merge.

18_NFC_MODULE / 成品读卡模块接口

NFC候选：Seeed Grove NFC 113020006 / PN532，立创历史料号C445568；现货待核实。

J201：JST SH 1.0mm C160404；1=SCL 2=SDA 3=3.3V 4=GND；须定制线束，非Qwiic针序。



PN532七位I2C地址0x24；CAP1296地址0x28，避免原RFID2同地址冲突。

本版NFC接口改为3.3V供电；不得接要求5V输入的M5Stack RFID2。

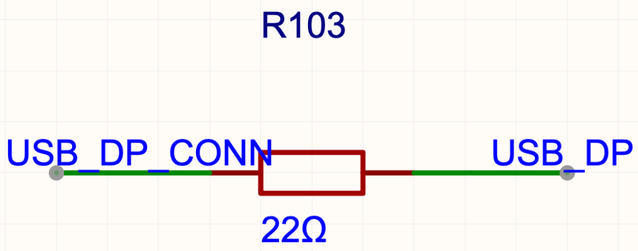
Grove NFC默认UART，使用前按厂家说明切换到I2C；100kHz起调，轮询响应。

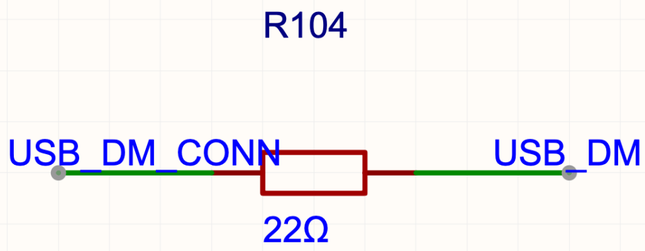
模块与天线外置；主板只提供接口。装壳核实天线空间、读卡距离和线束针序。

线端：SHR-04V-S-B + SSH-003T-P0.2-H；PCB背面右侧向外插接；不可直接使用原2.0mm插头。

原理图	Spirit_Native	创建日期	2026-10-24
		更新日期	2026-10-24
板子	01_主控板_Main	图页	18_NFC_MODULE
绘制		Spirit_V080_Three_Board_Layout_20261026	
审阅			
		版本	尺寸	页 12 共 13
嘉立创EDA		V1.0	A4	嘉立创EDA

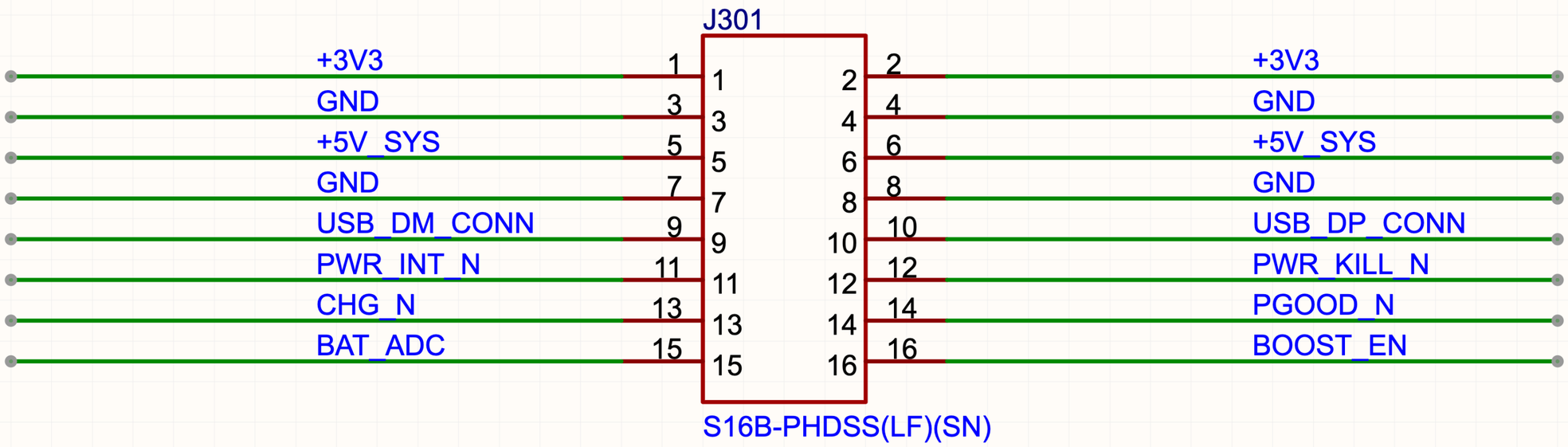
USB 串联 22Ω 电阻留在主控板，布局靠近 ESP32-S3。





V080 主控板 <-> 电源板 / 16-pin power + USB + control

J301 JST S16B-PHDSS(LF)(SN) / C161675 / 2.0mm 2x8



线束 H1：PHDR-16VS 两端，按针号 1->1 ... 16->16；禁止镜像接线。

1/2=3V3；5/6=5V；3/4/7/8=GND。电源/地用 AWG24，信号用 AWG28。

9/10 USB D-/D+ 用 90Ω 差分双绞线；首版线束长度目标 <=100mm。

只在断电时插拔；线束固定并远离 ESP32 天线区域；本版本仅布局，未布线。